

# Principles of Business Analytics

## Project



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# 1. Business Scenario

**Business Case:** Expansion into Emerging Markets

**Client:** Cloud-9 Solutions

**Sector:** Enterprise Software and Digital Services

## **Business Overview:**

Cloud-9 Solutions is a mid-sized tech company offering cloud-based software that allows clients to manage operations and workflows online, providing anytime, anywhere access with automatic updates and reduced IT costs.

## **Business Scenario:**

Having established strong growth in its home market, Cloud-9 Solutions is exploring expansion into emerging markets such as Africa. These regions are attractive due to their large populations, rapid digital adoption, and growing business sectors.

## **Business Question:**

Which emerging markets should Cloud-9 prioritize for expansion, based on the availability of digital infrastructure (e.g. internet, electricity), economic stability (e.g. GDP) and labour market conditions (e.g. urbanization, unemployment)?

## **Current business challenge:**

Cloud-9's team faces difficulties managing and analysing data from multiple international sources, with indicators like GDP growth, internet penetration, electricity availability, inflation, and unemployment. Datasets are often inconsistent, incomplete, and time-consuming to integrate, requiring manual spreadsheet work that is inefficient, error-prone, and hard to reproduce. To address this, Cloud-9 engaged Phi Consulting to design a structured relational database, consolidate data, resolve inconsistencies, and generate actionable insights for expansion strategy.

By centralizing these indicators, Cloud-9 can:

- Compare countries systematically across economic, infrastructure, and labour dimensions.
- Generate a ranked shortlist of expansion candidates based on objective criteria.
- Identify risks (e.g., unreliable electricity) before committing resources.
- Improve reporting efficiency through automated updates.

This approach enables faster, more reliable decisions, reduces the risk of entering unsuitable markets, and supports long-term, evidence-based operational planning.

## 2. Data collection and preparation

To support Cloud 9's expansion analysis, publicly available, raw datasets were sourced. Publicly available datasets were sourced from Kaggle and other open platforms to support Cloud-9's expansion analysis. The datasets included economic, infrastructure, and labour indicators relevant to market assessment, such as GDP, internet penetration, electricity access, urbanization, and unemployment.

The raw data was processed, cleaned, and integrated into a structured relational database, forming the foundation for the ETL (Extract, Transform, Load) process. Once loaded, the database enabled SQL-based analysis and the generation of actionable, data-driven insights to identify suitable emerging markets for expansion.

## 3. Relational Schema Design

Based on Cloud 9's business scenario and the sourced datasets, a relational schema was developed to organize country-level economic, infrastructure, and labour market indicators effectively. The schema is designed to:

- Reflect real-world relationships between countries and their various indicators.
- Support efficient queries across multiple dimensions (economic, infrastructure, labour, political).
- Enable scalable analysis as new data becomes available.

## 4. SQL-Based Analytical Procedures for Market Expansion

To guide Cloud-9's expansion into emerging markets, a series of SQL-based analytical procedures were executed on the relational database. These procedures systematically evaluate candidate countries across multiple dimensions critical to market readiness: economic performance, digital infrastructure and labour market conditions.

Each step builds upon the previous one, progressively narrowing the field of potential African markets and highlighting actionable insights. Steps 1–4 focus on data reliability, economic growth, GDP per capita, and infrastructure, producing an initial shortlist of countries. Steps 5–6 then assess labour availability, urbanization, and composite market attractiveness, providing a comprehensive framework for identifying the most promising expansion opportunities.

By applying this structured SQL-driven approach, Cloud-9 can ensure that its expansion decisions are evidence-based, risk-informed, and operationally feasible.

## Step 1: Data Completeness Check – Excluding Countries with Missing GDP

**Business Question:** Are the datasets sufficiently complete across candidate countries?

**Insight:**

Some countries had missing GDP data. Attempting to impute using regional averages led to unrealistic results, overestimating GDP compared to neighbouring high-performing economies.

**Business Value:**

GDP is a critical indicator for evaluating market size and economic stability. Countries with missing GDP data were excluded from further analysis to prevent misleading recommendations and focusing Cloud-9's expansion efforts on markets with trustworthy data.

## Step 2: Identify High-Growth Countries

**Business Question:** Which African countries exhibit the highest GDP growth between 2024 and 2025, and how does growth relate to GDP per capita?

**SQL Approach:**

Calculate  $(\text{GDP}_{2025} - \text{GDP}_{2024}) / \text{GDP}_{2024} * 100$  for each country and display alongside GDP per capita.

**Insight:**

Countries such as **Burundi, Malawi, and Uganda** show the highest growth rates (>10%) but very low GDP per capita.

**Business Value:**

This step provides an initial ranking of fast-growing economies and flags early that high growth alone does not guarantee a viable market for Cloud-9's cloud-based services.

### Step 3: Filter Countries by GDP per Capita

**Business Question:** Which African countries have sufficient GDP per capita to support adoption of Cloud-9's services, regardless of recent growth rates?

**SQL Approach:**

Select all countries and apply a GDP per capita threshold to identify markets with sufficient wealth. Include additional indicators, such as electricity and internet access, to assess operational feasibility.

**Insight:**

While **Seychelles, Mauritius, and Gabon** scored highly on economic measures such as GDP per capita and growth, only Seychelles combines strong wealth with high internet and electricity penetration. Mauritius and Gabon, despite their economic strengths, lag several countries outside the top three in infrastructure readiness. This demonstrates that GDP per capita alone is insufficient, as robust electricity and internet access are critical for technology-driven market expansion.

**Business Value:**

This step focuses on market readiness rather than raw growth, identifies infrastructure gaps early, and enables risk-informed expansion into markets with the highest likelihood of successful technology adoption.

### Step 4: Infrastructure Screening – Electricity and Internet Access

**Business Question:** Which high-GDP-per-capita countries also have sufficient electricity and internet access?

**SQL Approach:**

Filter candidates by electricity ( $\geq 95\%$ ) and internet penetration ( $\geq 65\%$ ).

**Insight:** Morocco, Mauritius, and Tunisia have strong infrastructure, ensuring Cloud-9's services can be reliably deployed. Other high-GDP countries may lack sufficient connectivity.

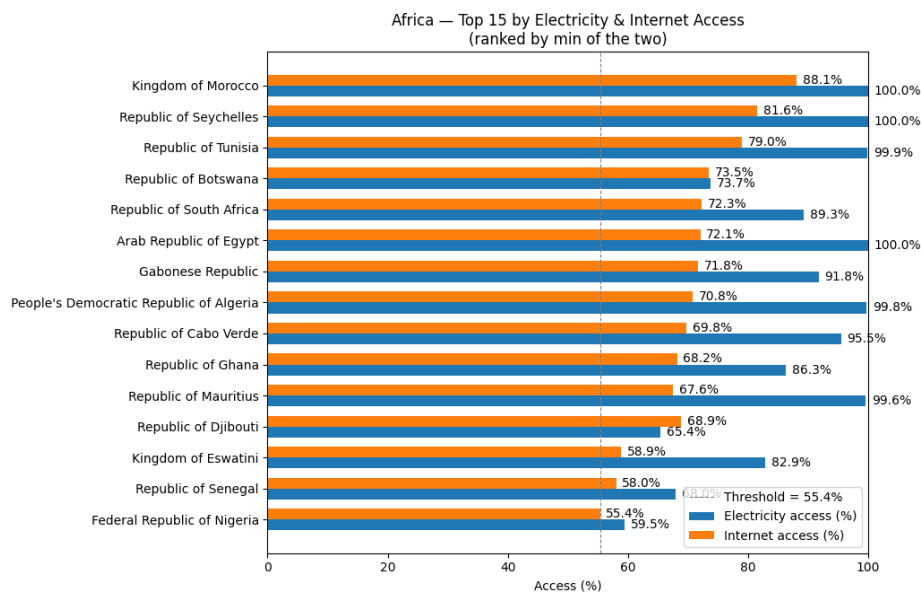


Figure 1: Africa – Top 15 by Electricity & Internet Access

**Business Value:** Confirms operational feasibility, reducing deployment risk.

## Step 5: Unemployment Analysis – Workforce Availability and Labour Conditions

**Business Question:** Which candidate countries have labour markets that could support Cloud-9's operations?

### SQL Approach:

We analysed unemployment rates for the six shortlisted countries from Step 4: Morocco (MA), Tunisia (TN), Mauritius (MU), Cape Verde (CV), Algeria (DZ), and Seychelles (SC). Each country was ranked to assess labour availability, providing insight into recruitment potential and market readiness.

**Insight:**

- Mauritius (7.23%) and Seychelles (7.94%) show the lowest unemployment rates, indicating tight labour markets with potentially higher hiring competition and wage pressure. The environments may favour specialised or high-skill recruitment but limit ease of rapid scale-up.
- Morocco (10.49%) and Algeria (11.55%) provide larger available workforces. However, Morocco's GDP growth (6.75%) enhances attractiveness as an investment destination, while Algeria's sluggish growth (1.5%) raises caution.
- Cape Verde (13.57%) and Tunisia (16.12%) present the most accessible labour pools. While this could support large-scale hiring, it may also reflect structural unemployment and skills mismatch, requiring training investments or partnerships with local institutions.

**Business Value:** Unemployment indicators are core to location analysis for operations strategy. Higher unemployment can indicate an available but underutilised workforce, presenting opportunities for economic impact and favourable labour conditions (ILO 2020; World Bank 2023).

This step allows Cloud-9 to align recruitment with labour dynamics, assess talent acquisition potential, and reduce operational risk through workforce-ready environments.

## Step 6: Land Use Analysis – Urban Space vs. Population Density

**Business Question:** How do shortlisted African countries compare in terms of urban land availability per person and urban population density, and what does this mean for Cloud-9's operational strategy?

**SQL Approach:**

Two key metrics were used: Urban m<sup>2</sup> per Person (a proxy for space availability) and Urban People per km<sup>2</sup> (a proxy for urban density). We quantified urban space availability by dividing urban land area by population, with the urban density metric used to verify overall urban crowding levels.



**Insight:**

- Mauritius has the highest urban land per person (377 m<sup>2</sup>) and the lowest urban density (2,653.5/km<sup>2</sup>), making it the most spacious and potentially logistically efficient.
- Tunisia strikes a strong balance with 233 m<sup>2</sup> per person and moderate density (4,293.4/km<sup>2</sup>), indicating a well-managed urban spread.
- Morocco, though denser (8,105.4/km<sup>2</sup>), maintains a reasonable 123 m<sup>2</sup> per person, supported by strong GDP growth (6.75%).
- Seychelles has very high urban density (7,485.3/km<sup>2</sup>) and only 134 m<sup>2</sup> per person, limiting scalability despite digital strengths.
- Algeria shows slightly better space (208 m<sup>2</sup>) but poor GDP growth (1.5%) which disincentivises prioritisation at this stage.

**Business Value:**

Urban density and land per capita are widely used indicators in infrastructure planning and urban economics. Studies show that cities with high land availability per person tend to support better logistics, lower real estate costs, and improved quality of life, key for long-term business operations (Angel et al. 2011; UN-Habitat 2020).

By focusing on these metrics, Cloud-9 can assess urban scalability, avoid overcrowded markets, and align its physical footprint with population and economic patterns.

## 5. Concluding remarks and Strategic Recommendation

Based on the SQL-based analytical procedures and the relational database analysis, Mauritius (MU), Tunisia (TN), and Morocco (MA) emerged as the most viable markets for Cloud-9's expansion. These countries combine sufficient GDP per capita, robust electricity and internet infrastructure, and manageable labour market conditions, making them operationally feasible and strategically attractive.

**Recommendation:**

Among the three, **Mauritius** presents the strongest overall opportunity for immediate expansion. It offers a high GDP per capita (\$12,000), excellent electricity access (99.6%), and adequate internet penetration (67%), alongside a moderate growth rate (3.62%) and a balanced labour market. While Seychelles has higher wealth per capita, its smaller population limits scale, and Morocco offers strong infrastructure but lower GDP per capita relative to Mauritius.

Focusing initial expansion efforts on Mauritius allows Cloud-9 to deploy its cloud-based software efficiently, mitigate operational risks, and capture a growing, digitally connected market. Seychelles and Morocco remain secondary opportunities for phased expansion as the company scales its operations in the region.

## 6. Appendix

### 1. Data Tables and Sources

- a. Raw datasets sourced from Kaggle and other open platforms.
  - i. Indicators include GDP (2024, 2025), GDP per capita, GDP growth rate, internet penetration, electricity access, unemployment, inflation, land use, and governance type.
- b. Links or references to the original datasets:  
(<https://github.com/chantelle1402/MSBA-SQL-Queries>)  
(<https://github.com/chantelle1402/MSBA-SQL-Queries/tree/main/1.%20Data%20Tables%20%26%20Sources>)

### 2. Relational Schema (link)

- a. Diagram of the PostgreSQL relational database schema.
- b. Description of tables, primary keys, foreign keys, and relationships.  
(<https://github.com/chantelle1402/MSBA-SQL-Queries/tree/main/2.%20Relational%20Schema>)

### 3. SQL Code

- a. Full SQL queries executed for Steps 1–6:  
(<https://github.com/chantelle1402/MSBA-SQL-Queries/tree/main/3.%20SQL%20%26%20Python%20Code>)
- b. Includes queries for filtering by GDP growth, GDP per capita, infrastructure, unemployment, and land use.

### 4. Additional Tables and Charts

- a. Extended country-level data for Africa and other regions.
- b. Chart visualizing internet/electricity access.  
(<https://github.com/chantelle1402/MSBA-SQL-Queries/tree/main/4.%20Additional%20Tables%20%26%20Charts>)

## 7. Citations

1. International Labour Organisation (ILO). 2020. *World Employment and Social Outlook: Trends 2020*. Geneva: International Labour Organisation.
2. World Bank. 2023. *World Development Indicators: Labour Force Data*. Washington, DC: World Bank Group.
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4. United Nations Human Settlements Programme (UN-Habitat). 2020. *World Cities Report 2020: The Value of Sustainable Urbanization*. Nairobi: United Nations Human Settlements Programme.